```
‘’’
LP-III Assignment 6 (Group A - DAA)
Write a program for analysis of quick sort by using deterministic and randomized variant.
Name: Tanmay Sandip Bothara
Roll No: 23 Batch: BE_Comp_A1
‘’’

import random
import timeit

def partition(arr, low, high, randomized=False):
    if randomized:
        p = random.randint(low, high)
        arr[p], arr[high] = arr[high], arr[p]

    pivot = arr[high]
    i = low - 1

    for j in range(low, high):
        if arr[j] < pivot:
            i += 1
            arr[i], arr[j] = arr[j], arr[i]

    arr[i + 1], arr[high] = arr[high], arr[i + 1]
    return i + 1


def quick_sort(arr, low, high, randomized=False):
    if low < high:
        p = partition(arr, low, high, randomized)
```

```
        quick_sort(arr, low, p - 1, randomized)
        quick_sort(arr, p + 1, high, randomized)


if __name__ == "__main__":
    n = int(input("Enter number of elements: "))

    if n <= 0:
        print("Please enter a positive number.")
    else:
        # Generate unique random elements
        arr = random.sample(range(1, n * 10 + 1), n)

        # Measure deterministic Quick Sort
        det_time = timeit.timeit(
            lambda: quick_sort(arr.copy(), 0, n - 1, False),
            number=1
        )

        # Measure randomized Quick Sort
        rand_time = timeit.timeit(
            lambda: quick_sort(arr.copy(), 0, n - 1, True),
            number=1
        )

        print("\nArray size:", n)
        print(f"Deterministic Quick Sort: {det_time:.6f} seconds")
        print(f"Randomized Quick Sort:    {rand_time:.6f} seconds")
```

‘’’

OUTPUT:-

Enter number of elements: 1000

Array size: 1000

Deterministic Quick Sort: 0.001195 seconds

Randomized Quick Sort: 0.001358 seconds

‘’’